Advanced Design Research 2 (2024) 1–13

Contents lists available at ScienceDirect

Advanced Design Research

journal homepage: www.keaipublishing.com/IJADR

Design of plant protection unmanned aerial vehicles: Quantitative research and virtual simulation

Bin Qi, Shuhang Zhao *, Xiaoming Sun **, Li Wang

College of Agricultural Engineering and Food Science, Shandong University of Technology, Zibo, 255049, China

ARTICLE INFO

Keywords:
Kansei engineering
Virtual simulation
Modeling design
Plant protection UAV
Fluid mechanics

ABSTRACT

To enhance people's perceptions of agricultural machinery and elevate agricultural equipment to a high-end status, this paper proposes a product design evaluation process based on quantitative theory and virtual simulation technology. First, an "imagery–part library–virtual simulation" product design model is established through the integration of Kansei engineering and artificial intelligence (AI) drawing technology to construct a virtual part library. Assembly experiments are then conducted using a virtual simulation platform. Subsequently, a combined evaluation model, based on fuzzy comprehensive evaluation and behavioral experiments, is developed to evaluate and screen plant protection unmanned aerial vehicles (UAVs) from both the component and whole-system perspectives. Finally, the final scheme is simulated and optimized from the perspective of aerodynamics. This paper analyzes the modeling of plant protection UAVs from aesthetic and engineering viewpoints. It quantifies aesthetic requirements through various methods and integrates cutting-edge AI-assisted design and virtual simulation technology to optimize the initial product design process and consequently reduce the subjectivity of the product modeling process. The proposed methodology holds both theoretical and practical significance for the modeling design of agricultural machinery equipment.

1. Introduction

China is a leading agricultural country with over 1.8 billion mu of arable land. Urbanization has led to a significant increase in the loss of the agricultural labor force and in the demand for intelligent agricultural machinery [1]. Plant protection unmanned aerial vehicles (UAVs) stand out among various intelligent equipment owing to their high efficiency, convenient operation, and affordability, making them an essential tool in promoting agricultural mechanization [2]. Changing young people's perception of agricultural machinery is essential to encourage them to join the agricultural industry. With the maturity of the industry, consumer demands for functionality decrease, while the demand for aesthetic appeal increases [3]. While functional requirements are the priority during the early development stages of plant protection UAVs, a focus on aesthetic design is required as the technology matures. Thus, the current plant protection UAV modeling design stage is pivotal in promoting a balance between product appearance and functionality.

"Form follows function" is a critical principle in product design, where product modeling heavily relies on the designer's experience and understanding of user needs, demanding high proficiency. Researchers have attempted to enhance the aesthetic value of products using quantitative theory [4–6] and mathematical models [7,8]. As early as 1995, Japanese scholar Nagamachi pointed out that Kansei engineering is a new type of consumer-oriented ergonomics product development technology and divided Kansei engineering into three categories, which provided help for designers in the fields of automobile, clothing, and electrical appliances [9]. Some researchers also compensate for the deficiency of a quantitative method in product design by combining multiple quantitative methods. Rampal obtained consumers' feelings about electric vehicles using the Kano model and employed Quality Function Deployment (QFD) to identify priority areas for improving product configurations. The established model can help electric vehicle manufacturers evaluate and improve their services, thereby improving customer satisfaction [10]. To enhance the efficiency of current product concept generation and evaluation methods, Fazeli proposed a product design and evaluation method based on axiomatic design and QFD, and the effectiveness of the process was proven through a case study involving a hand rehabilitation device [11]. Additionally, fuzzy mathematics theory has been widely employed in product design and evaluation. Zhao used an unclear hierarchy comprehensive evaluation method

* Corresponding author.
** Corresponding author.
E-mail addresses: shsx1997@163.com (S. Zhao), 1966377001@qq.com (X. Sun).

https://doi.org/10.1016/j.ijadr.2024.03.001
Received 11 December 2023; Received in revised form 11 March 2024; Accepted 12 March 2024
Available online 20 April 2024
2949-7825/© 2024 Northwestern Polytechnical University. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

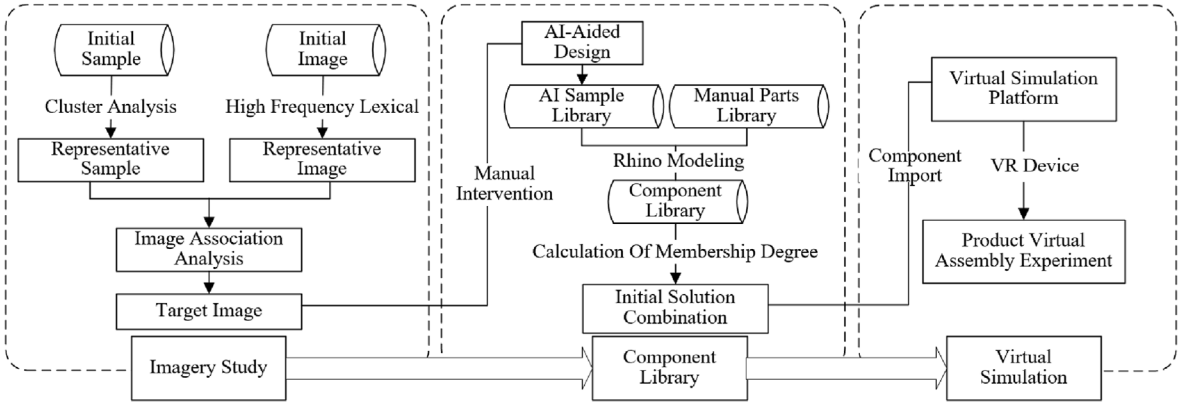


Fig. 1. Product modeling design evaluation model based on quantitative theory and virtual simulation.

to evaluate the ergonomics of plant protection UAV remote control and optimized the results using Theory of Inventive Problem Solving(TRIZ) theory [12]. Quantitative theory has emerged as a prevalent approach in product appearance and function design. Current research emphasizes the acquisition of enhanced design outcomes through the amalgamation of quantitative methods or the application of quantitative methodologies from diverse disciplines to product design.

Virtual simulation experiments utilize virtual reality (VR) technology to simulate manufacturing processes and operational effects in realistic scenarios. Initially employed in military and scientific research, virtual simulation has become more accessible owing to technological advancements and cost reductions. It is now widely used in various industries, including intelligent manufacturing, education, and art design [13–16]. To enhance collaboration between users and designers, Bruno incorporated virtual simulation into the product evaluation process, enabling users to participate in the creation of the interfaces of products such as microwave ovens and washing machines [17]. Virtual simulation technology has been significantly advanced across diverse domains and reached a relatively mature state. Consequently, this technology adequately meets the requirements of virtual simulation systems for general industrial product design.

The product modeling approaches adopted in previous studies involved considerable time and labor costs [18–20]. The present study introduces artificial intelligence (AI)-assisted design, which allows for the production of a large number of target products in a short period based on imagery vocabulary, thereby improving design efficiency. Owing to the uncertainty associated with AI-assisted design, two assessment methods, the fuzzy comprehensive evaluation method and the behavioral experimentation method, are used in this study to ensure design accuracy.

The present study proposes a process for evaluating product modeling design, encompassing product modeling, program evaluation, and modeling optimization. This research primarily focuses on assessing the modeling design of plant protection UAVs. The study aims to develop a product design evaluation process that can be easily replicated and applied to evaluate shape design in other agricultural machinery and equipment.

2. Establishment of product modeling design evaluation model

The proposed imagery–part library–virtual simulation (IPV) design model based on Kansei engineering and virtual simulation is shown in Fig. 1.

2.1. Product image research

Step 1 Representative samples are obtained. First, the market of plant protection drones is analyzed in terms of form. Python is used to obtain relevant plant protection drone picture samples from the Internet. The initial pictures are processed and screened, and a similarity questionnaire is set for sample screening. The questionnaire results are then imported into SPSS software, and a systematic cluster analysis is conducted. Representative samples are finally obtained according to the effects of pedigree analysis.

Step 2 Representative images are obtained. First, a representative image is acquired. Imagery words are typically used to gather users' perceptions of products in Kansei engineering research [18]. First, numerous initial image words about the target product are collected through web crawlers (Python), brainstorming, and interviews. Subsequently, existing literature on Kansei

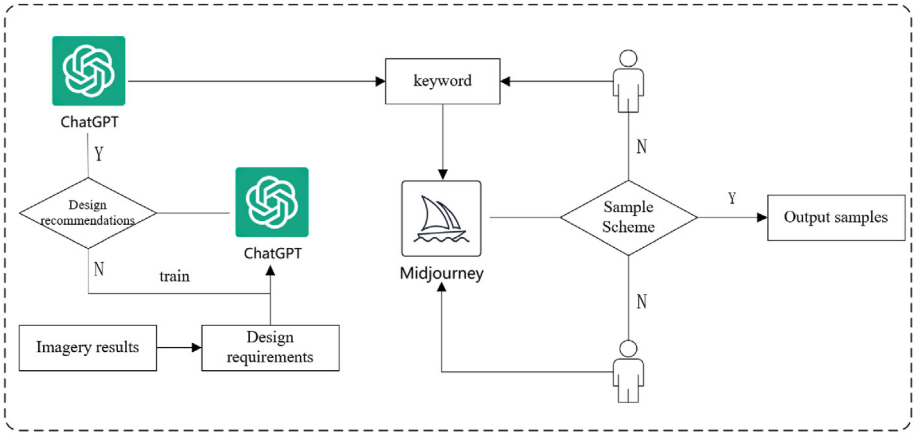


Fig. 2. AI-aided design operation process.

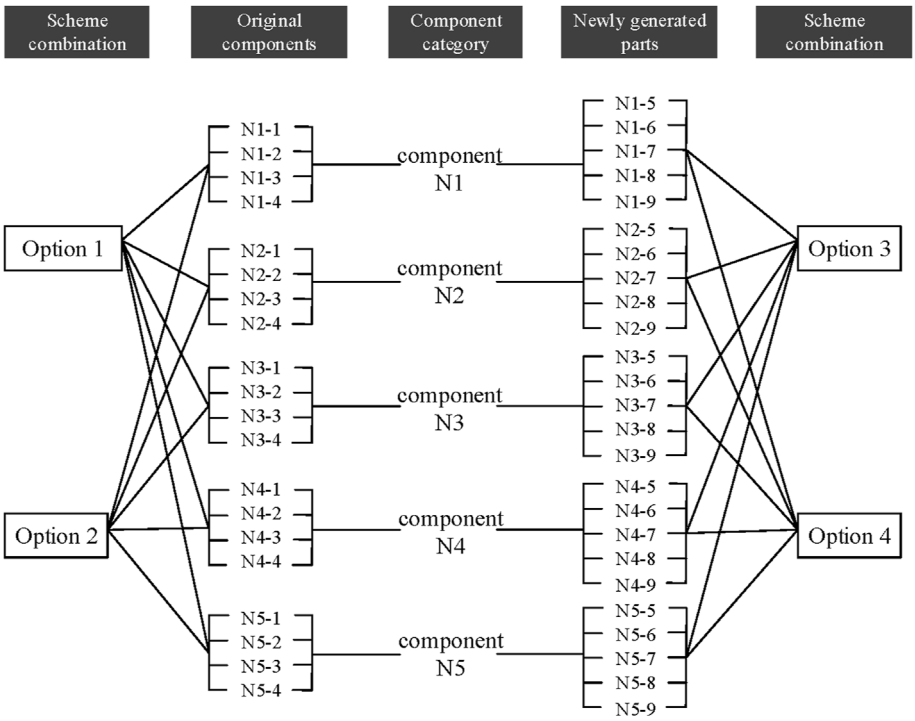


Fig. 3. Combination of initial design schemes.

engineering research is reviewed, and the usage of perceptual words is counted. Words that coincide with the initial image words of this study and are used more than three times are selected. These words are considered high-frequency image words, representing representative image words.

Step 3 Imagery correlations are analyzed. A correlation questionnaire between representative samples and initial vocabulary is established, and the scoring groups are defined. The top three images are designated as strong correlation images to guide the construction of the virtual part library for the target products below [19].

2.2. Building a library of virtual components

2.2.1. AI sample library construction

AI-assisted design involves the use of artificial intelligence technology to assist designers in completing design work. Currently, many designers have already employed AI-assisted design in product design-related fields. Midjourney is one of the more sophisticated AI-assisted drawing tools. Midjourney's mapping principles involve keyword control and graph generation. This study utilized the image analysis results and AI-assisted design tools (ChatGPT and Midjourney) to construct an AI sample library, as depicted in Fig. 2.

2.2.2. Establishment of virtual component library

According to the AI sample library, appropriate samples are selected to establish a virtual component library. The process involves dividing the samples based on their component requirements and developing a three-dimensional model using Rhino software. For each component, a total of nine elements are selected. Among them, elements 1 to 4 in the first component library represent the effects of the original products on the market, used for comparison with the newly designed plant protection drone combined in the component library to verify the feasibility of the design model. Elements 5 to 9 are from the AI component library.

2.3. Virtual assembly experiment

2.3.1. Initial scheme combination

Step 1 Component membership acquisition. The evaluation team scores the correlation between the shape of each component and the image vocabulary according to the seven-point Likert scale method (1 being completely unrelated, 7 being completely related) [20]. The membership degree Z calculation formula can be expressed as follows:

$$Z = \frac{\sum_{a=1}^{a=7} ab}{\sum_{c=1}^{c=9} \sum_{a=1}^{a=7} ab} \tag{1}$$

where a denotes the evaluation score, b denotes the number of this score, and c denotes the component element number.

Step 2 Initial scheme combination. The modeling design scheme of the target product is generated from each of the old and new part libraries, as shown in Fig. 3. The standard for generating schemes involves selecting combinations with the highest and lowest membership degrees from these libraries. There are four initial design schemes.

2.3.2. Virtual assembly experiment

Step 1 A virtual simulation assembly experiment platform for target products is established using Unity3D, and the initial scheme components are imported into the assembly experiment scene. Additionally, the User Interface(UI) is designed.

Step 2 According to the initial scheme combination determined by membership degree, users can assemble and design the target product using a mouse or a handle. The completed products can be fully displayed to the user by adjusting the perspective. Additionally, simulation experiments can be conducted in the virtual simulation system for products with special requirements.

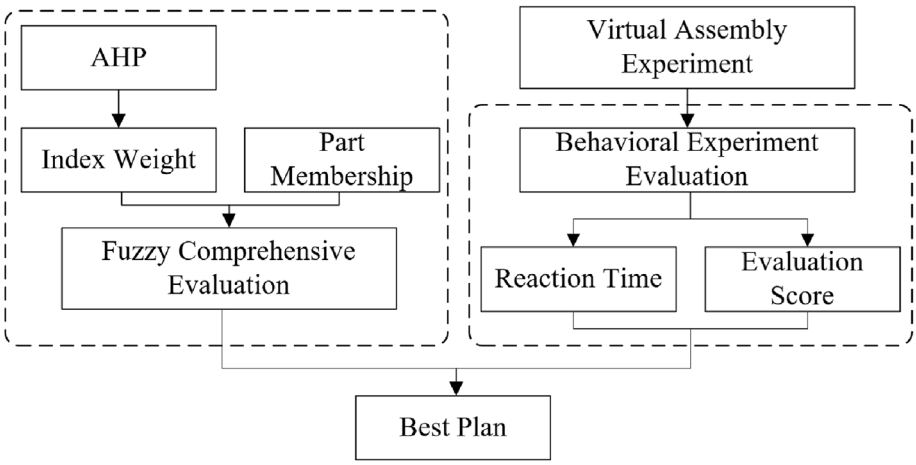


Fig. 4. Combined evaluation model.

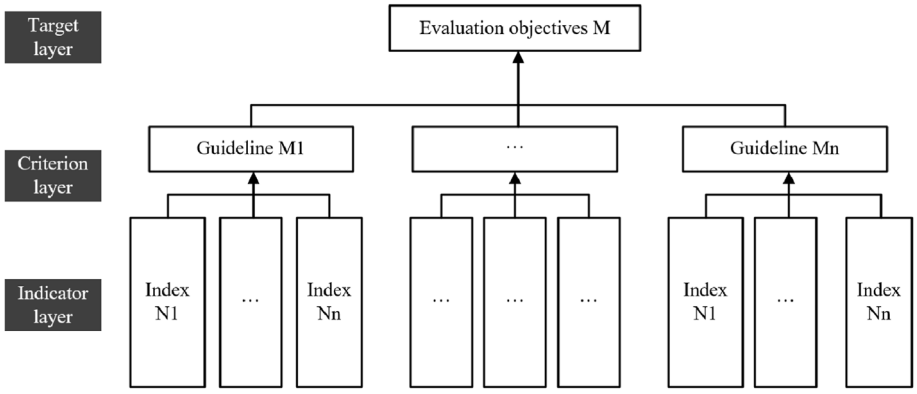


Fig. 5. Hierarchical model.

(2) Calculation of weight factor (*W*) through the analytic hierarchy process.

3. Establishment of a combined evaluation model

A combined evaluation model based on fuzzy comprehensive evaluation and behavioral experiments is established, as shown in Fig. 4. A fuzzy comprehensive evaluation involves the calculation of the memberships and weights of different components from the component dimension. In contrast, behavioral experiment evaluation involves the comprehensive assessment of the results of virtual simulation experiments at an overall level. The feasibility of this study's product modeling design model is assessed through the combination of these two evaluation methods.

3.1. Fuzzy comprehensive evaluation

The fuzzy comprehensive evaluation method is used to assess uncertain indicators through fuzzy mathematics[21]. The core part of the method is the construction of a membership evaluation matrix and the determination of the index weight coefficient. The construction of the matrix requires the designing of a scientific comment set to satisfy the user. The weight coefficient determination generally combines the analytic hierarchy process and the entropy weight method to increase the objectivity of the index weight. According to the analytical hierarchy process, this paper summarizes the specific steps of fuzzy comprehensive evaluation as follows:

(1) Construction of the fuzzy comprehensive evaluation index.

The hierarchical analysis of the target product is conducted, and the elements in the target layer, the criterion layer, and the index layer are taken as the evaluation indexes (Fig. 5).

The weight reflects the importance of each indicator in the evaluation system and influences the scientificity of the final evaluation result. This paper uses the Analytic Hierarchy Process(AHP) method to ascertain the weight of the evaluation system. The method is briefly summarized into the following two steps:

Step 1 Building a judgment matrix. The criteria layer and each element under the criteria layer are compared in pairs, and the comparison results are filled in the corresponding positions of the table. The scoring standard establishes an evaluation matrix based on the nine-point scale table.

Step 2 Calculating the index weight value. The evaluation matrix is calculated using the sum-product method to obtain the weight value *W* of each index. To prevent logical errors in the evaluation matrix, a consistency test is conducted on the matrix. CR < 0.1 indicates that the consistency of the matrix is satisfactory and that the test is passed. Otherwise, the matrix needs to be adjusted.

(3) Construction of the membership matrix (*R*)

The membership degrees of the four initial schemes are assembled into a membership degree matrix *R*.

(4) Outputting of fuzzy comprehensive evaluation results (*B*)

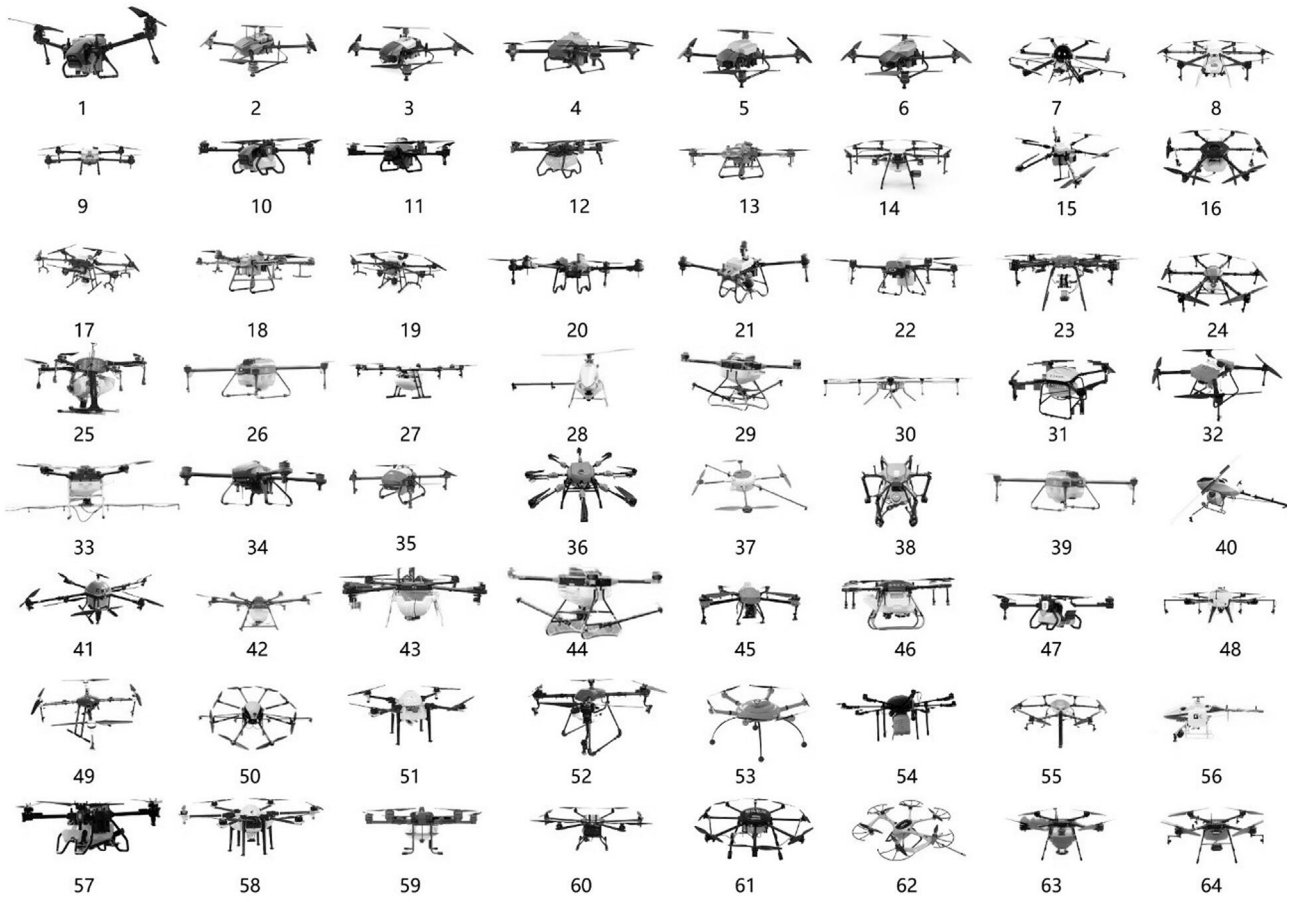


Fig. 6. Initial sample.

We combine the weight obtained through AHP with the membership degree matrix to obtain the evaluation score for the target product.

$$B = WR \tag{2}$$

3.2. Behavioral experiment evaluation

E-Prime is software jointly developed by Carnegie Mellon University, the University of Pittsburgh, and PSYCHOLOGY SOFTWARE TOOLS,INC (PST)in the United States for conducting behavioral experiments and data collection. It can accurately measure people's reaction time when fed with images, sounds, or texts[22]. The software includes a built-in programming module, enabling expert scoring, image search, and other experiments to be conducted by writing programs alone. The module can also serve as a stimulus source for completing related experiments involving eye movement, Electroencephalography(EEG) [18], and other

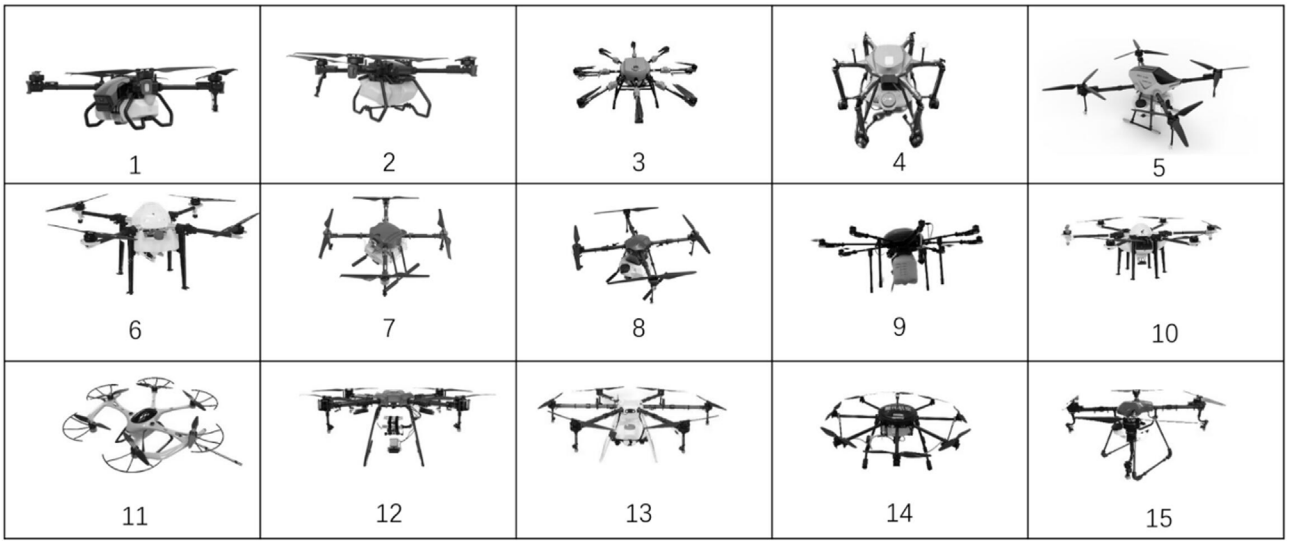


Fig. 7. Representative sample.

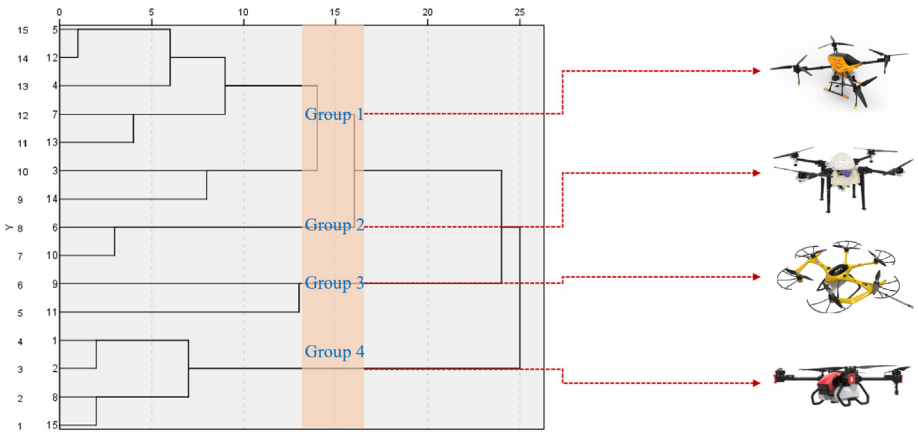


Fig. 8. Pedigree plot constructed through the mean join (between groups) method.

Step 2 Acquiring representative images. Through online retrieval of evaluations, brainstorming, interviews, and other methods related to plant protection UAV products, 80 image vocabularies of UAV modeling design are collected. Through the high-frequency image vocabulary method, 10 words suitable for the design of plant protection drones are selected from the current Kansei engineering literature as representative image vocabulary (Table 1).

equipment.

In this study, E-Prime is utilized to assess the initial scheme of the target product. Participants evaluate and score the scheme and image vocabulary, and their evaluation scores and reaction times are subsequently recorded. A virtual assembly experiment precedes the behavioral experiment to ensure that each participant fully comprehends the structure of the target product before conducting the behavioral experiment.

4. Modeling design and evaluation of plant protection UAV

4.1. Modeling design of plant protection drone

4.1.1. Imagery study

Step 1 Obtaining representative samples. According to previous market research, the multi-rotor plant protection UAV is easy to use, has a low failure rate, and is not difficult to promote. Particularly, the medicine box sinking UAV has a lower center of gravity and better stability. In this study, Python crawler technology is used to obtain sample pictures of plant protection drones. Sixty-four initial samples are obtained through preliminary screening (removing irrelevant, fuzzy, and conceptual machines), as shown in Fig. 6. Finally, 15 plant protection UAVs with representative medicine box sinking-type characteristics are manually screened (Fig. 7).

A similarity questionnaire is designed for the 15 UAVs. The questionnaire employs the five-point scale method (where 1 indicates very low similarity, and 5 indicates complete similarity), and 10 graduate students with five years of industrial design experience as expert members are selected as participants. The results of the similarity questionnaire are clustered and analyzed. After the samples are divided into four categories, the variation between the samples widens significantly. Therefore, the number of clusters is selected as 4. A representative sample is chosen from each category, resulting in the acquisition of four representative samples (Fig. 8).

Graduate students majoring in agricultural machinery and 20 undergraduates majoring in industrial design are assembled to form a scoring group. They evaluate the fit between the 10 initial vocabulary words and sample pictures. In the online scoring process, a seven-level scale method is used (−3 points means completely unrelated, 3 points means completely related). A total of 120 questionnaires are distributed (four questionnaires for each sample), and 104 valid questionnaires are recovered. The questionnaire results are summarized, and an image correlation line graph is generated according to the average image correlation degree table. Fig. 9 illustrates the image correlation for different samples, with the image correlation situation depicted on the right side.

Correlation analysis reveals that the three image vocabularies of atmosphere, technology, and simplicity are highly correlated with representative samples. Therefore, the images of atmosphere, technology, and simplicity are selected to guide the design of the plant protection drone.

4.1.2. Building a library of virtual components

According to the image analysis results, the design requirements for the plant protection drone entail an atmospheric, technological, and simple approach. First, the design requirements are transmitted to ChatGPT, and ChatGPT is repeatedly trained to provide design

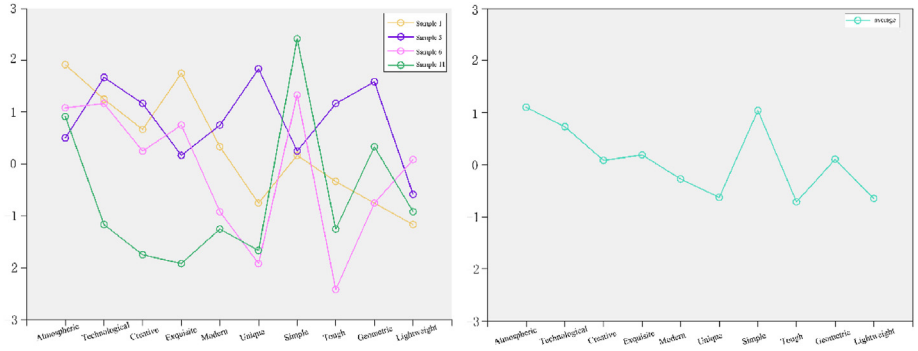


Fig. 9. Image association analysis line chart.



Fig. 10. AI sample library.

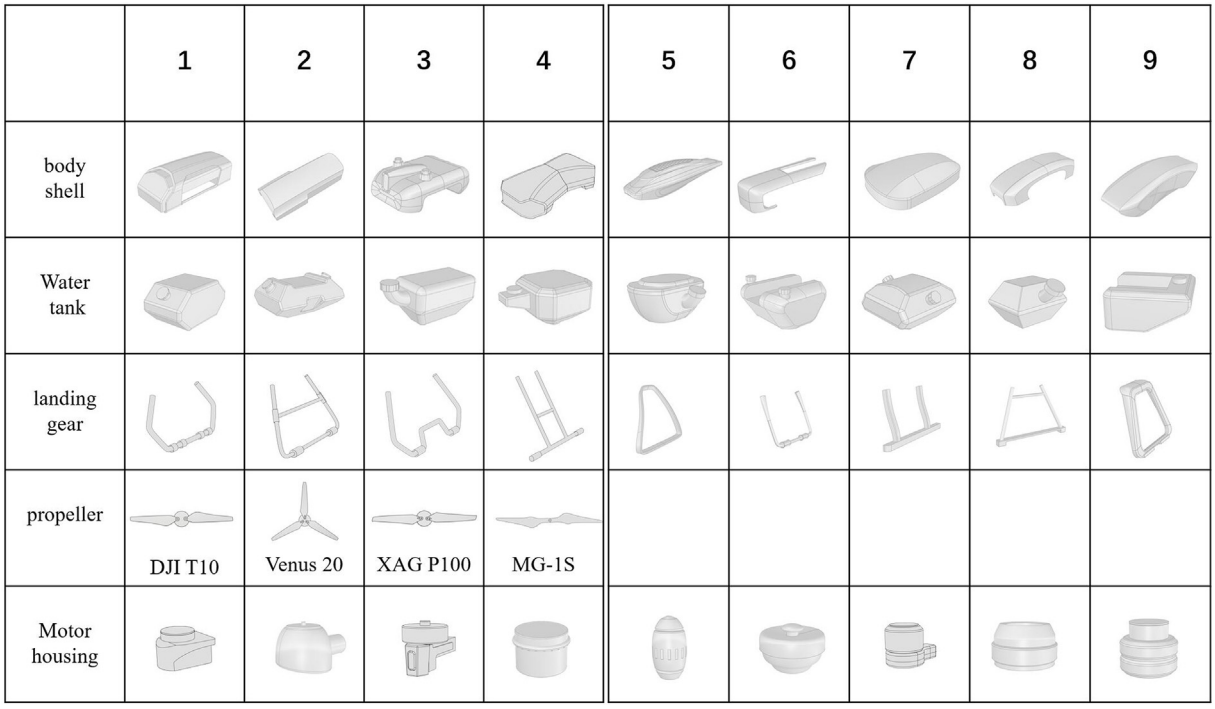


Fig. 11. Virtual component library.

suggestions and assist in filtering out keywords recognizable by AI tools. Afterward, keywords are input into Midjourney (such as agricultural drones, four propellers, water tank below, technological aesthetics, minimalist style, atmospheric appearance, white background, studio lighting, C4D, 4K resolution, chaos 50) to generate and display four sample scenarios at a time. Finally, after adjustments, 200 satisfactory sample plans are output, leading to the establishment of an AI sample library. A portion of the sample library is illustrated in Fig. 10.

With the AI-generated sample library as a reference, the five components of the plant protection drone—the body shell, medicine box, bracket, propeller, and motor shell—are visually rendered in three dimensions. This process establishes the virtual component library of the plant protection drone (Fig. 11). To ensure the flight feasibility of the plant protection UAV, the propeller parts in the library incorporate components from existing products (such as DJI T10, Venus 20, XAG P100, MG-1S), along with their associated motor components.

Table 1
Representative imagery vocabulary.

Number	1	2	3	4	5
Kansei words	Atmospheric	Technological	Creative	Exquisite	Modern
Number	6	7	8	9	10
Kansei words	Unique	Simple	Tough	Geometric	Lightweight

Table 2
Membership degree of component library.
(2) Initial plan

	Part 1	Part 2	Part 3	Part 4	Part 5	Part 6	Part 7	Part 8	Part 9
N1	0.0909	0.0994	0.0979	**0.1188**	**0.1246**	0.1157	0.1157	0.1149	0.1222
N2	0.1155	**0.1226**	0.0941	0.0995	0.1241	0.1015	**0.1276**	0.1034	0.1116
N3	**0.1312**	0.1021	0.1228	0.0866	**0.1320**	0.1021	0.1176	0.0934	0.1121
N4	0.2658	0.2227	**0.3061**	0.2053					
N5	**0.1150**	0.0974	0.0946	0.1084	0.1205	0.1139	**0.1225**	0.1131	0.1146

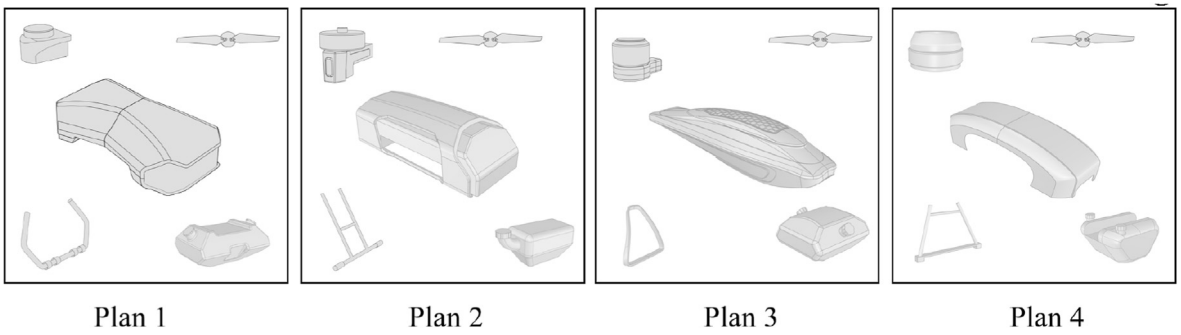


Fig. 12. Initial combination scheme.

(3) Scheme virtual assembly

4.1.3. *Virtual simulation experiment*

(1) Acquisition of membership degree

The evaluation team scores each element using a seven-point Likert scale. The scoring is based on the extent to which each element relates to the imagery vocabulary of atmosphere, technology, and simplicity (where 1 indicates entirely irrelevant and 7 denotes wholly related). The evaluation process combines both online and offline methods. The evaluation team comprises 20 graduate students majoring in agricultural machinery, 30 undergraduates majoring in industrial design, 5 project team members, 5 plant protection drone spraying personnel, and 5 agricultural machinery professional teachers, totaling 60 individuals. The evaluation results are tabulated and presented in Table 2 (the maximum values in the old and new databases are in bold, and the minimum values are underlined).



Fig. 13. Virtual assembly experiment.

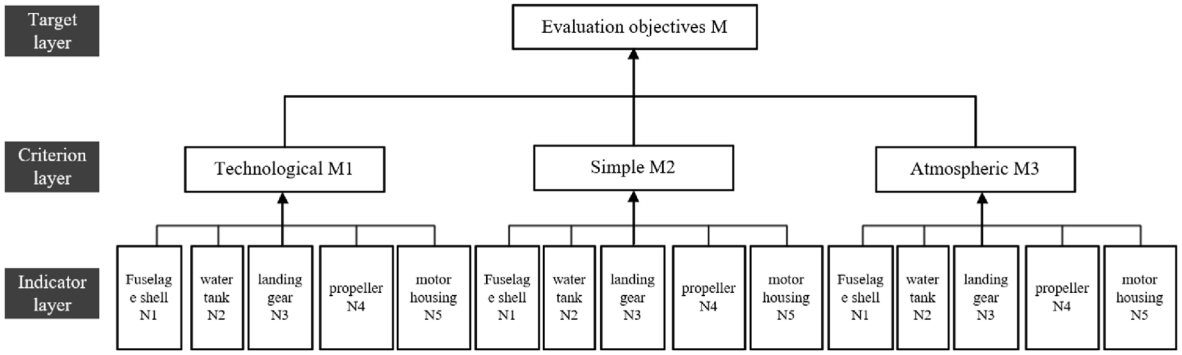


Fig. 14. Hierarchical structure model of plant protection UAV.

(2) Obtaining the weight vector

Two plant protection UAV solutions are generated from the old and new component libraries. According to the degree of membership, the components of the initial plan 1 are fuselage shell element 4, medicine box element 2, bracket element 1, propeller element 3, and motor shell element 1. The components of the initial plan 2 are fuselage shell element 1, medicine box element 3, bracket element 4, propeller element 4, and motor shell element 3. The components of the initial scheme 3 are fuselage shell element 5, medicine box element 7, bracket element 5, blade element 3, and motor shell element 7. The components of the initial scheme 4 are fuselage shell element 8, medicine box element 6, bracket element 8, paddle element 4, and motor shell element 8. The four schemes are shown in Fig. 12.

The virtual assembly experiment carried out in this study is based on the existing virtual simulation platform for agricultural drones in the laboratory. At this stage, assembly design tasks are assigned to the participants. The participants completed the plant protection drone assembly experiment using the VR device by manipulating the control handle (Fig. 13). The PICO neo3 pro is used as the VR device. As this study is a designated target experiment, participants are required to conduct the assembly experiment based on the combination of the four provided schemes. They are encouraged to feel the details of each component during assembly and experience the panoramic display of the final plan.

4.2. Evaluation of plant protection UAV solution

4.2.1. Comprehensive evaluation of fuzzy hierarchy

(1) Constructing the fuzzy comprehensive evaluation index

In this paper, the appearance of the plant protection drone is taken as the target layer of the hierarchical structure model, with the three perceptual images forming the criterion layer. The five secondary units—fuselage, medicine box, propeller, bracket, and motor casing—are divided into the index layer, as depicted in Fig. 14.

Ten agricultural machinery graduate students and 10 industrial design undergraduates (with a male-to-female ratio of 1:1) are invited to form a scoring focus group. The composition standard of the focus group is as follows: the group has members with agricultural machinery backgrounds and an understanding of plant protection drones and members with industrial design backgrounds and strong aesthetic abilities. Students majoring in agricultural machinery understand the structure and function of plant protection drones but need further development in aesthetics. Students majoring in industrial design have certain aesthetic qualities but need an understanding of the structure and function of plant protection drones. Therefore, the group ensures that the students can offer complementary perspectives during discussions.

First, the initial judgment matrix is obtained by consulting relevant literature on plant protection drones and Kansei engineering. Team members then express their opinions on the data in its initial form. The time allocated for each critical data discussion is 2 min, and a total of 33 data points need to be discussed, resulting in a total of 66 min. Finally, a

Table 3
Judgment matrix (M).

M	M_1	M_2	M_3	W_0
M_1	1	4	3	0.6232
M_2	1/4	1	1/2	0.1373
M_3	1/3	2	1	0.2395

Table 4
Judgment matrix (M1).

M1	N1	N2	N3	N4	N5	W_1
N1	1	5	7	8	3	0.5045
N2	1/5	1	3	5	1/3	0.1408
N3	1/7	1/3	1	5	1/3	0.0923
N4	1/8	1/5	1/5	1	1/5	0.0374
N5	1/3	3	3	5	1	0.2250

Table 5
Judgment matrix (M2).

M2	N1	N2	N3	N4	N5	W_2
N1	1	3	5	6	4	0.4693
N2	1/3	1	2	5	3	0.2312
N3	1/5	1/2	1	2	1/2	0.0910
N4	1/6	1/5	1/2	1	1/6	0.0484
N5	1/4	1/3	2	6	1	0.1601

Table 6
Judgment matrix (M3).

M3	N1	N2	N3	N4	N5	W_3
N1	1	5	6	7	3	0.4967
N2	1/5	1	2	4	1/4	0.1249
N3	1/6	1/2	1	2	1/3	0.0784
N4	1/7	1/4	1/2	1	1/5	0.0470
N5	1/3	4	3	5	1	0.2530

Table 7
Consistency ratio results.

	M	M1	M2	M3
λ_{max}	3.0182	5.2008	5.1452	5.1322
CI	0.0091	0.1004	0.0726	0.0566
CR	0.0176	0.0896	0.0648	0.0505

Table 8
Comprehensive weight.
(3) Constructing the membership matrix

	M1	M2	M3	*W*
N1	0.5045	0.4693	0.4967	0.4978
N2	0.1408	0.2312	0.1249	0.1494
N3	0.0923	0.0910	0.0784	0.0888
N4	0.0374	0.0484	0.0470	0.0412
N5	0.2250	0.1601	0.2530	0.2228

Table 9
Membership degrees of four schemes.

	Plan 1	Plan 2	Plan 3	Plan 4
N1	**0.1188**	0.0909	**0.1246**	0.1149
N2	**0.1226**	0.0941	**0.1276**	0.1015
N3	**0.1312**	0.0866	**0.1320**	0.0934
N4	**0.3061**	0.2053	**0.3061**	0.2053
N5	**0.1150**	0.0946	**0.1225**	0.1131

criterion-level judgment matrix (Table 3) and three index-level judgment matrices (Tables 4–8) are generated.

The consistency of the matrix is checked, as indicated in Table 7. CR is less than 0.1, indicating that the evaluation matrix is reasonable and has no logic error.

The comprehensive weight W of each component element in the index layer is calculated, as shown in Table 8. The weight set can be expressed as

$$W = (0.4987\ ,\ 0.1494\ ,\ 0.0888\ ,\ 0.0412\ ,\ 0.2228)$$

The membership degrees of the four initial schemes are summarized in Table 9.

The membership matrix R can be expressed as

$$\boldsymbol{R} = \begin{bmatrix} 0.1188 & 0.0909 & 0.1246 & 0.1149 \\ 0.1226 & 0.0941 & 0.1276 & 0.1015 \\ 0.1312 & 0.0866 & 0.1320 & 0.0934 \\ 0.3061 & 0.2053 & 0.3061 & 0.2053 \\ 0.1150 & 0.0946 & 0.1225 & 0.1131 \end{bmatrix}$$

(4) Fuzzy hierarchical comprehensive evaluation results

$$B = WR = (0.1281, 0.0973, 0.1336, 0.1153)$$

Therefore, the accuracy of the initial schemes of the four plant protection UAVs is in the following order: scheme 3 > scheme 1 > scheme 4 > scheme 2.

4.2.2. Behavioral experiment evaluation

A program for the modeling behavior experiment of plant protection UAVs is developed in E-Prime. Twelve image-scheme pictures are created for the four plant protection UAV schemes to serve as visual stimuli for the study. Users can evaluate the proposals by clicking the left mouse button.

The experimental procedure is outlined as follows:

(1) Participants complete the assembly experiments of the four initial UAV schemes using VR equipment.
(2) Participants first read the introductory instructions to understand the content of the behavioral experiment. After reading, they click the mouse to proceed to the experiment.
(3) Participants focus on the "+" sign displayed in the center of the screen for 1000 ms.
(4) Upon entering the evaluation interface, participants rate the degree of fit between the displayed vocabulary and the plan. They use a five-point scale method (1 indicating a poor match between the image and the plan, and 5 indicating a strong match). Participants click the corresponding score to proceed to the next interface.
(5) All content on the screen disappears, and participants can choose to take a short break or continue scoring the next scheme. Each participant needs to score 12 (3 × 4) pictures. The end of the experiment is indicated once the last set of ratings is completed. The specific process is illustrated in Fig. 15.

Considering that the test content focuses on the correlation between the scheme and image vocabulary, it is unrelated to the professional background of the participants. Hence, a total of 20 graduate students and undergraduates from various majors are recruited for this behavioral experiment. Following the experiment, 240 pieces of experimental data are collected and statistically analyzed (Fig. 16).

Fig. 16 reveals that the image "simple" exhibits the shortest reaction time in schemes 1, 2, and 4, suggesting that it is easier to evaluate compared to other images. Across these three schemes, "simple" exhibits the most distinct feature. However, the image "atmospheric" elicits longer reaction times across all four schemes, suggesting that the image features are less prominent, requiring participants to spend more time making judgments. The total average reaction times of the four schemes increase in the following order: scheme 3 < scheme 2 < scheme 4 < scheme 1. These results show that the image features of schemes 2 and 3 are more prominent.

Analysis of the behavioral experiment results reveals that the response times required for the evaluation of schemes 3 and 2 are relatively short; however, both schemes exhibit the highest and lowest

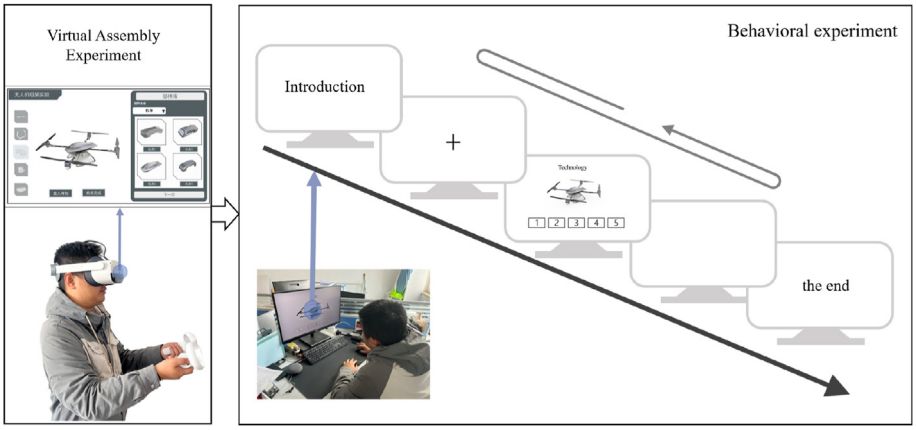


Fig. 15. Behavioral experiment operation flow.

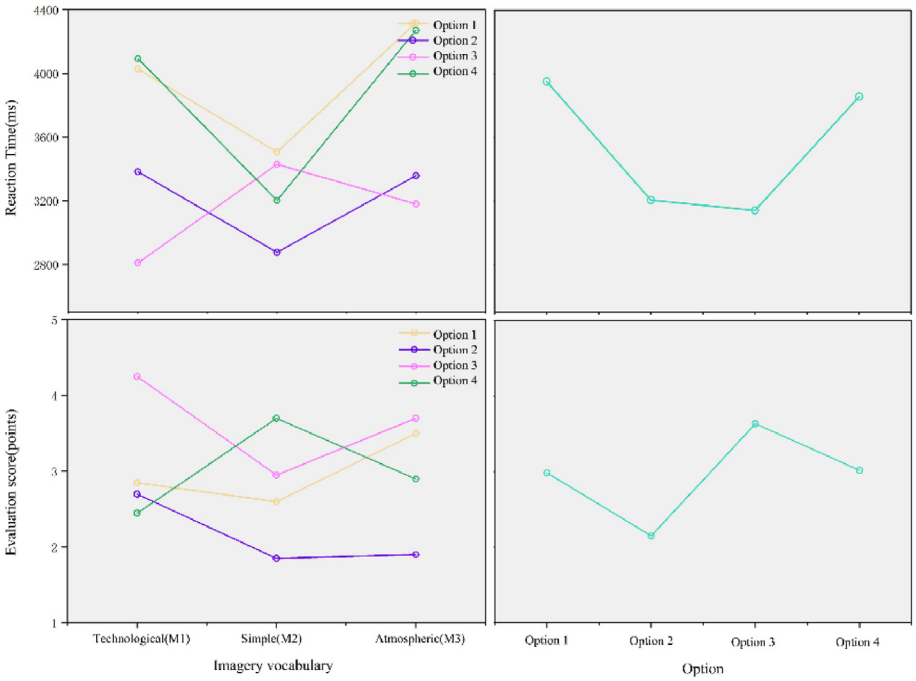


Fig. 16. Relationship between imagery and the average evaluation score (left).

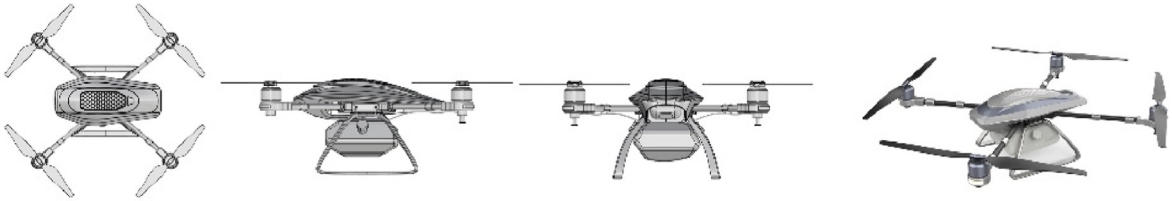


Fig. 17. Final design plan.

evaluation scores, respectively. Therefore, the image features of schemes 3 and 2 are the most apparent. Although scheme 1 has a higher evaluation score, it requires the longest evaluation reaction time. Hence, the imagery feature of this scheme is the vaguest.

4.2.3. *Final design scheme output*

A combination of the fuzzy comprehensive evaluation results and the behavioral experiment results reveals that the component scores and overall scores of scheme 3 surpass those of other schemes, and the imagery characteristics of scheme 3 are also the most prominent. Therefore, scheme 3 is chosen as the final design scheme, as shown in Fig. 17.

5. Scheme optimization and expression

5.1. *Aerodynamic analysis of plant protection UAV solution*

The multi-rotor plant protection UAV is affected by not only natural wind resistance during operation but also the complex wind field generated by the rotor interacting with the UAV body. Therefore, studying the wind field of plant protection drones is crucial. Ansys software offers a dedicated fluid dynamics analysis module. This analysis is conducted using Ansys Fluent to complete the aerodynamic analysis of the plant protection drone scheme.

Owing to the complex surface and numerous rounded corners of the model in scheme 3, meshing cannot be achieved. First, the model is

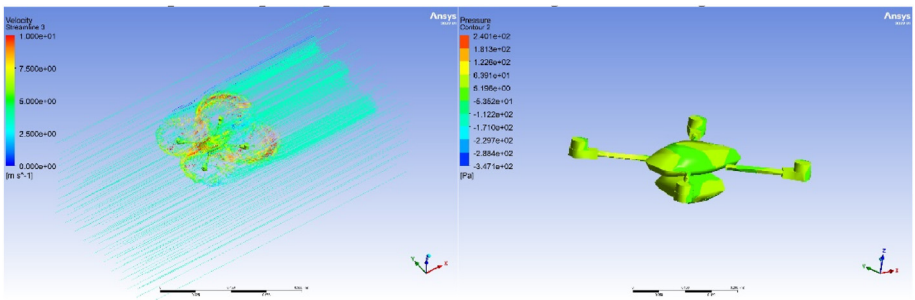


Fig. 18. Wind field distribution and airframe pressure distribution of plant protection UAV.

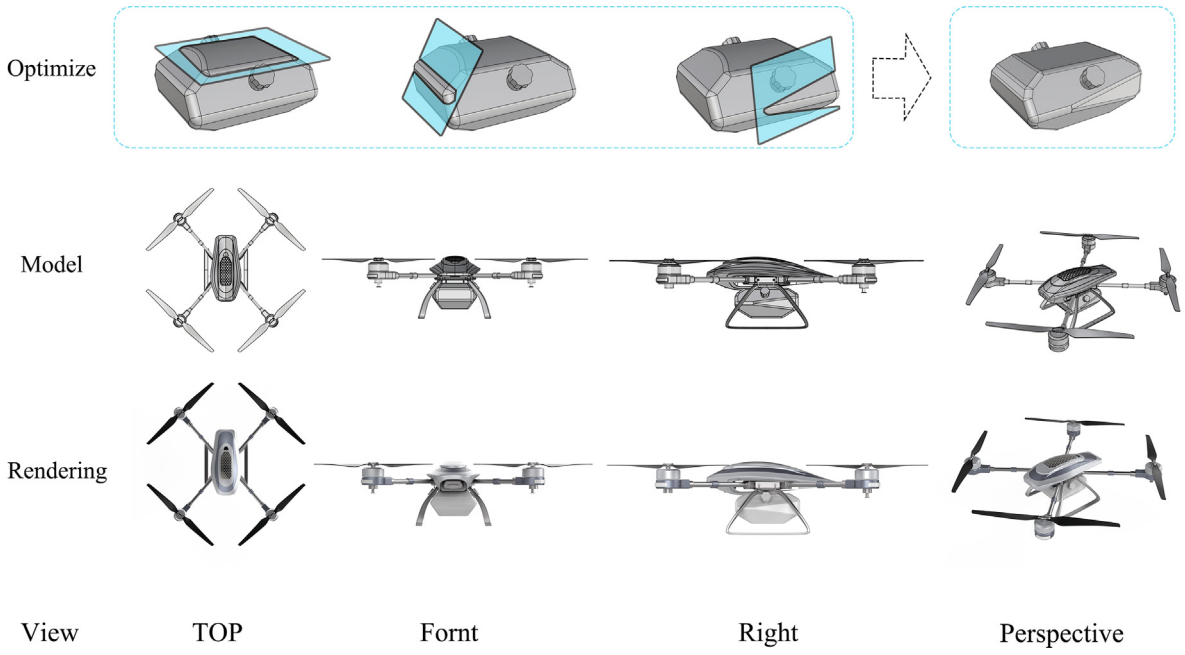


Fig. 19. Scheme optimization process.

simplified. The brackets, fillets, holes, and small internal parts are removed to ensure the integrity of the main body. Afterward, the size of the plant protection drone is specified. The drone's unfolded dimensions are 2487 mm × 2460 mm × 700 mm, and the XAG P4718 foldable propeller (diameter 47 inches × pitch 18 inches) is utilized. The model is imported into Fluent for meshing. During the solution process, the wind speed is set to 3 m/s, and the propeller speed is set to 3800 rmp according to the specifications of XAG P100. Finally, the wind field distribution and body pressure cloud map of the plant protection UAV are generated (Fig. 18).

According to the wind pressure distribution on the fuselage, several areas exhibit high pressure around the medicine box. Specifically, the high pressure in front of the medicine box results from the combined effect of the propeller and natural wind, while the rear of the medicine box experiences pressure due to the wind field generated by the rotation of the two rear propellers. Additionally, slightly higher pressure exists on the top and front of the fuselage shell.

5.2. Shape optimization design

We simplify the model by removing all rounded corners and complex surfaces. In the original model, the fuselage shell exhibits high fluidity; therefore, the fuselage is not optimized separately. However, according to the pressure distribution, numerous high-pressure areas are identified within the medicine box, indicating a significant issue with its shape. Therefore, an optimized design for the shape of the medicine box is necessary. The medicine box is optimized while ensuring that its visual characteristics remain unchanged. The optimization focuses on refining the sides and front of the box to reduce areas of high gloss and rear pressure and converging the shapes of the side and rear portions. Fig. 19 displays the optimization process.

6. Conclusion

The established IPV modeling design integrates Kansei engineering with AI-assisted design. AI-assisted design enables the expression of design based on imagery outcomes, which addresses the issue of inaccuracies resulting from designers' subjective interpretations of imagery. The virtual simulation assembly experiment replaces the previous single-rendering display.

Virtual simulation technology provides new ideas for product design. This study employs the plant protection UAV virtual simulation system to construct a plant protection UAV component library, and a virtual simulation assembly experiment is conducted. The virtual simulation replicates an actual behavioral scenario, thereby increasing the accuracy of the experimental results.

This paper evaluates the plan of plant protection UAVs from both local and holistic perspectives through the fuzzy comprehensive evaluation method and the behavioral experiment method. The consistent results of the evaluations validate the efficacy of the "shaping design-scheme evaluation" model established in this paper. Subsequently, the modeling scheme of the plant protection UAV is optimized from an aerodynamic standpoint. Hydrodynamic simulation of the design scheme reveals that both the UAV fuselage and the medicine box are affected by the natural wind and the wind field generated by the rotors. Consequently, localized optimization of the medicine box area, which is subjected to higher pressure, is conducted, and the resulting design scheme is effectively demonstrated.

Ethical statement

This study has no ethical implications.

Funding

This research received no external funding.

Author contributions

Conceptualization, B.Q., and S.Z.; data curation, S.Z.; investigation, L.W., X.S. and S.Z.; method-ology, S.Z., and X.S.; formal analysis, B.Q.; project administration, B.Q., C.H.; supervision, X.S.; writing—original draft, S.Z.; writing—review and editing, B.Q., and X.S.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The author acknowledges that this work has been supported by the Key research and development project of Shandong Province (project number: 2021LYXT012) and the independent innovation special project of Shandong Province - Intelligent Laser Grader Research and Industrial Demonstration (project number: 2013CXC90203). We also thank Grammarly (https://app.grammarly.com/) for the language assistance provided during the writing of this manuscript. The authors thank TopEdit (www.topeditsci.com) for its linguistic assistance during the preparation of this manuscript.

References

[1] Y. Zhang, H. Long, S. Chen, L. Ma, M. Gan, The development of multifunctional agriculture in farming regions of China: convergence or divergence? Land Use Pol. 127 (2023) 106576.
[2] H. Li, H. Zhu, Z. Jiang, Y. Lan, B. Engineering, Performance characterization on downwash flow and spray drift of multirotor unmanned agricultural aircraft system based on CFD, Int. J. Agri. 15 (3) (2022) 1–8.
[3] M. Ding, Z. Bai, Application. Product color emotional design adaptive to product shape feature variation, Color Res. 44 (5) (2019) 811–823.
[4] M. Liu, C. He, Jianqiang Zou, Form design of agricultural equipment based on virtual Kansei engineering, Packag. Eng. (6) (2022) 43.
[5] Y. Cheng, X. Xu, G. Chen, Evaluation of engineering vehicle modeling design based on analytic hierarchy process and Kansei engineering, Packag. Eng. 41 (20) (2020) 85–90.
[6] Y. Wang, Y. Ke, K. Wang, J. Guo, F. Qin, A composition-oriented aesthetic view recommendation network supervised by the simplified golden ratio theory, Expert Syst. Appl. 195 (2022) 116500.
[7] J.M. Sur, D.J. Kim, Logistics. Comprehensive risk estimation of maritime accident using fuzzy evaluation method–Focusing on fishing vessel accident in Korean waters, Asian J. Shipping 36 (3) (2020) 127–135.
[8] S.T. Koçak, F. Yercan, Comparative cost-effectiveness analysis of arctic and international shipping routes: a fuzzy analytic hierarchy process, Transport Pol. 114 (2021) 147–164.
[9] M. Nagamachi, Kansei Engineering: a new ergonomic consumer-oriented technology for product development, Int. J. Ind. Ergon. 15 (1) (1995) 3–11.
[10] A. Rampal, A. Mehra, R. Singh, A. Yadav, K. Nath, A.S. Chauhan, Kano and QFD analyses for autonomous electric car: design for enhancing customer contentment, Mater. Today: Proc. 62 (2022) 1481–1488.
[11] H.R. Fazeli, Q. Peng, Generation and evaluation of product concepts by integrating extended axiomatic design, quality function deployment and design structure matrix, Adv. Eng. Inf. 54 (2022) 101716.
[12] S. Zhao, B. Qi, L. Wang, Y. Li, H. Zhu, R. Gao, Human-machine evaluation and optimization of plant protection UAV remote control based on AHP/FCA/TRIZ, J. Shandong Univ. Technol. 36 (6) (2022) 46–51.
[13] K. Hagita, Y. Kodama, M. Takada, Simplified virtual reality training system for radiation shielding and measurement in nuclear engineering, Prog. Nucl. Energy 118 (2020) 103127.
[14] T. Fujimoto, Y.W. Park, Balancing supply chain competitiveness and robustness through "virtual dual sourcing": lessons from the Great East Japan Earthquake, Int. J. Prod. Econ. 147 (2014) 429–436.
[15] The development of the virtual testing system of the self-driving cleaning truck based on virtual reality technology, in: H. Yang, J. Wu, X. Wang, C. Gao (Eds.), Journal of Physics: Conference Series, IOP Publishing, 2019.
[16] D. Manca, R. Totaro, S. Nazir, S. Brambilla, S. Colombo, Virtual and augmented reality as viable tools to train industrial operators. Computer Aided Chemical Engineering vol. 31, Elsevier, 2012, pp. 825–829.
[17] F. Bruno, M. Muzzupappa, Product interface design: a participatory approach based on virtual reality, Int. J. Hum. Comput. Stud. 68 (5) (2010) 254–269.
[18] Y. Yang, D. Chen, S.-H. Yu, M. Yu, B. Qi, Pan-community product form design based on shape grammar, Comput. Integr. Manuf. Syst. 19 (9) (2013) 2107–2115.
[19] M. Liu, C. He, J. Zou, Design of agricultural equipment form based on virtual perceptual engineering, Packag. Eng. 43 (6) (2022) 41–48.
[20] Y. Yang, H. Tian, S. Jiao, X. Tian, Image-driven product design based on function-behavior-structure, Mech. Design Manuf. (6) (2022) 176–179.